

Permutations vs Combinations

Date_____ Period____

State if each scenario involves a permutation or a combination.

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| 1) Mofor and Darryl are planning trips to ten countries this year. There are 14 countries they would like to visit. They are deciding which countries to skip. | 2) A team of 13 basketball players needs to choose a captain and co-captain. |
| 3) The batting order for ten players on a 11 person team. | 4) A group of 35 people are going to run a race. The top 7 finishers advance to the finals. |

State if each scenario involves a permutation or a combination. Then find the number of possibilities.

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| 5) Selecting which seven players will be in the batting order on a 8 person team. | 6) There are 15 applicants for two Manager positions. |
| 7) You are setting the combination on a five-digit lock. You want to use the numbers 62413 but don't care what order they are in. | 8) 3 out of 15 students will ride in a car instead of a van |
| 9) A team of 12 dodgeball players needs to choose a captain and co-captain. | 10) There are 60 people at a meeting. They each give a Valentine's Day card to everyone else. How many cards were given? |
| 11) There are 30 students at a meeting. They each give a Valentine's Day card to everyone else. How many cards were given? | 12) Carlos has homework assignments in six subjects. He only has time to do four of them. |
| 13) A group of 20 people are going to run a race. The top three runners earn gold, silver, and bronze medals. | 14) There are 190 politicians at a meeting. They each shake hands with everyone else. How many handshakes were there? |